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Exploratory Analysis and Predictive Modeling of Quarterback Performance Data

Introduction:

The quarterback position is one of the most scrutinized and important positions in all American football. It is often regarded as the only position whose individual success is based on team success or failure. Evaluating a quarterbacks (qb's) career can be difficult as it involves a variety of performance statistics, team success, and accolades. This report aims to explore a dataset of historical National Football Leagues (NFL) qb's statistics and uncover what exactly makes a qb successful, and how that success translates into the way we view those players today, mostly when it comes to Hall of Fame (HOF) status.

The dataset used in this analysis was sourced from Pro Football Reference (PFR), a commonly used database for all kinds of NFL statistics for all positions including quarterbacks. This dataset was used to get data on passing yards, touchdowns, interceptions, passer rating, and all other columns of data used in this report. All these statistics are commonly used to determine qb success. This dataset helps allow for descriptive and predictive analysis to find conclusions about performance, correlation, and career achievements.

Understanding quarterback statistics like passing yards, touchdowns, wins, and passer rating among others and how they relate to Hall of Fame status is valuable for fans and analysts to be able to properly describe the most important attributes and accolades players receive. For example, the comparison of statistics of HOF quarterbacks to qbs who haven't been inducted shows the standard of greatness between the two. When using a dataset like the one created, all of the players supplied are incredible NFL players, so finding the differences between HOF and non-HOF players is key to find what makes a player and all-time great. Also, exploring stats like interceptions affect passer rating can help quantify how much mistakes weigh on production.

Additionally, analyzing relationships between draft year and wins, we can try to understand whether changes through time in the NFL, offensive styles, and player development have influenced quarterback success through eras. This can help sport historians determine how to look at the different eras of qb's and how much their statistics mean.

Ultimately, this analysis contributes to the wider field of sports analytics by providing statistic led insights into player evaluation and legacy building in football. By blending statistical techniques, descriptive statistics, correlation analysis, regression modeling, and visualization, we aim to answer three key questions:

1. What statistical differences exist between Hall of Fame and non-Hall of Fame quarterbacks?
2. How do interceptions and touchdowns respectively influence a quarterback's passer rating?
3. Has the era in which a quarterback was drafted affected their total career wins?

These questions will help add context to past performances and influence how current and future can be evaluated by fans and experts alike.

Methods:

Dataset Description:

The dataset used in this report for analysis is a quarterback statistical dataset created by the author of this paper. The statistics are sourced from *Pro Football Reference* (PFR) and are up to date as the statistics were sourced after the last football season ended. A few key observations to keep in mind about these datasets are as follows: many younger players will not be represented fairly as they haven't had the time played that the retired players have, and the older players generally have worse statistics despite having HOF status due to the era they played in.

The columns of data used are:

- “name” – The players name.
- “teams” – Teams the player has been on.
- “passing_yards” – A continuous column of data used to show how many yards a qb has thrown the ball for.
- “tds” – A continuous column of data used to show how many touchdowns a qb has thrown the ball for.
- “ints” - A continuous column of data used to show how many interceptions a qb has thrown the ball for.
- “passer_rating” - A continuous column of data used to show how a qb's statistics best reflect how well they have played.
- “numb” – The players number.
- “wins” – A continuous column of data of the players career wins.

- “superbowl” – A continuous column of data of how many superbowl the player has won.
- “hof” – A categorical column of data 1 means they made the hall of fame, and 0 means they haven’t.
- “year_drafted” – The year a player was drafted, may be used categorical to find different eras of players.

Tools and Software:

Python was chosen as the primary programming language for this analysis due to its readability, versatility, and the large variety of data science libraries it offers. Its syntax makes it accessible for beginners, while being a commonly used language ensures extensive documentation for extended research. Several Python libraries were used for data manipulation, visualization, statistical analysis, and machine learning.

The panda’s library was used to load and manage the quarterback dataset, allowing for efficient data wrangling tasks such as filtering, cleaning, and computing summary statistics. For data visualization, matplotlib.pyplot provided tools for generating scatter plots, bar charts, and other essential graphs that helped identify trends and relationships within the data. The scipy library supported statistical testing, including functions like the two-sample Kolmogorov-Smirnov (KS) test, which was used to compare the distributions of continuous variables between different player groups.

To implement the machine learning component, the scikit-learn library played a central role. The KNeighborsClassifier was employed to perform a k-nearest neighbors classification, aiming to predict Hall of Fame status based on player statistics. To evaluate model performance,

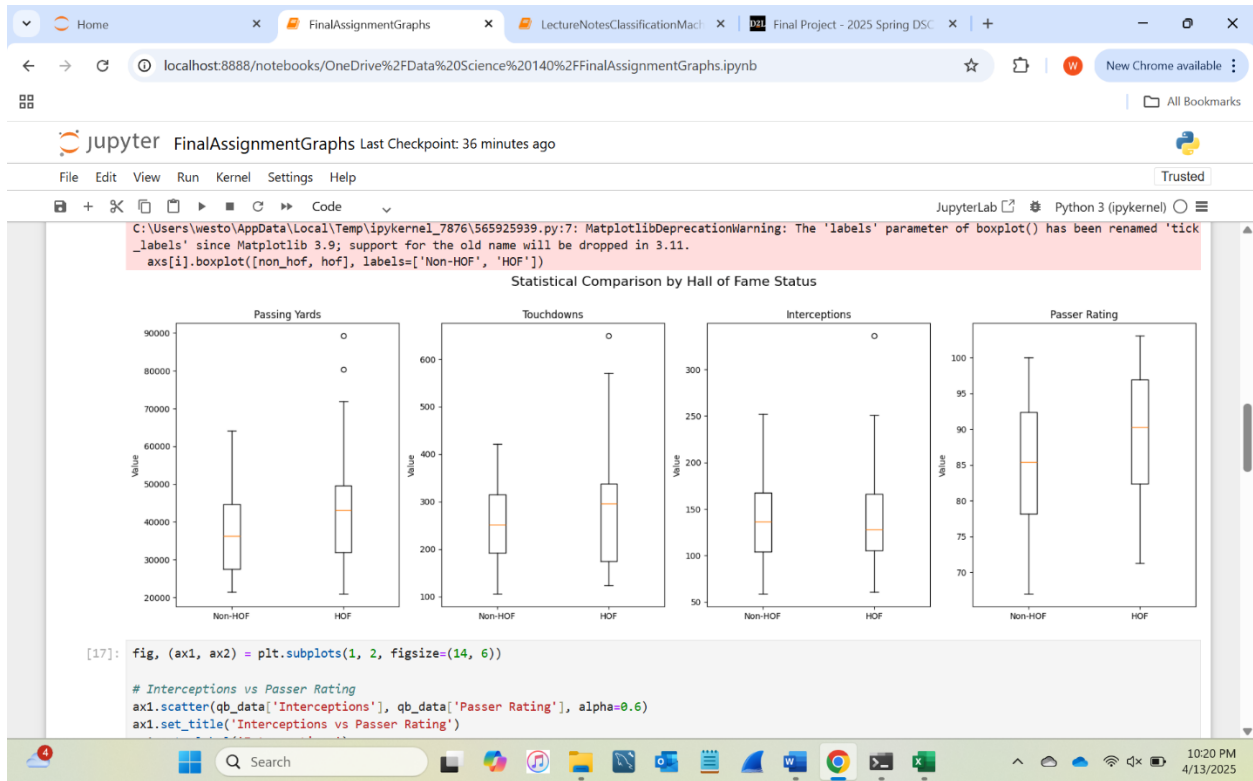
the data was split into training and testing subsets using `train_test_split`, ensuring that model accuracy was measured on previously unseen data. The `accuracy_score` function was used to quantify prediction correctness, and `ConfusionMatrixDisplay` provided a visual breakdown of true and false predictions across classes. Together, these tools allowed for a comprehensive and replicable analysis of quarterback performance data.

Analytical Techniques:

A combination of statistical and machine learning techniques was applied to analyze the quarterback dataset. Descriptive statistics, including mean, range, and standard deviation, were computed for continuous variables, while frequency counts were used to summarize categorical variables. Relationships between continuous variables were investigated through scatter plots, linear regression analysis, and calculation of Pearson and Spearman correlation coefficients. To compare the distributions of continuous data across different groups, a two-sample Kolmogorov-Smirnov (KS) test was performed. Associations between categorical variables were evaluated using Chi-Squared tests, particularly to assess relationships such as Super Bowl wins and Hall of Fame induction. For predictive modeling, a k-nearest neighbors (KNN) classifier was implemented, with model performance assessed using accuracy scores and confusion matrix visualization. Additionally, feature importance analysis and principal component analysis (PCA) were employed to extend the analysis and provide further insight into the most influential variables and to reduce the dimensionality of the dataset.

Results:

Figure 1: Statistical Comparisons by HOF status

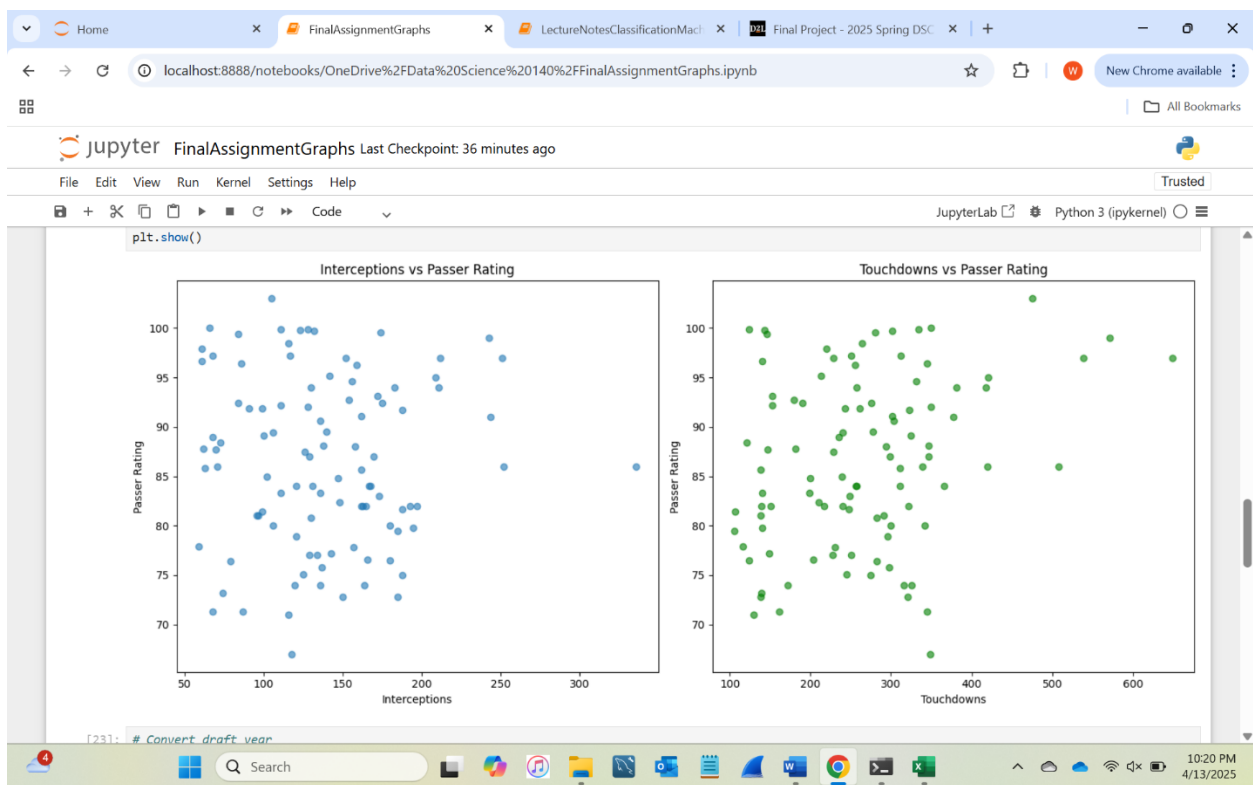


This figure shows the comparison of Hall of Fame qb's vs non-Hall of Fame qb's statistics. Due to this dataset being a comparison of the most accomplished qb's ever this is worthy to look at as they must all have similar statistics. When looking at the positive statistics (Yards, Touchdowns, Passer rating) we can see that the HOF qb's generally have a higher median, ceiling, lower floor, and a larger variability. This is what is expected, as if you're in the HOF you probably have been very successful statistically. The one negative statistic of interceptions is generally the same with one much larger outlier. This is probably since the best players get more opportunities to play longer and therefore will have more

opportunities to make mistakes as well. However, this shows that the difference between a great non-HOF player and a HOF player is not the mistakes they make but the great plays they make instead.

From a more objective stand point, independent sample t-tests confirmed that these differences were statistically significant. The difference in passer rating ($t \approx 4.75$, $p < 0.001$) and touchdown count ($t \approx 4.10$, $p < 0.001$) were particularly strong, suggesting that these variables play an important role in distinguishing Hall of Fame players.

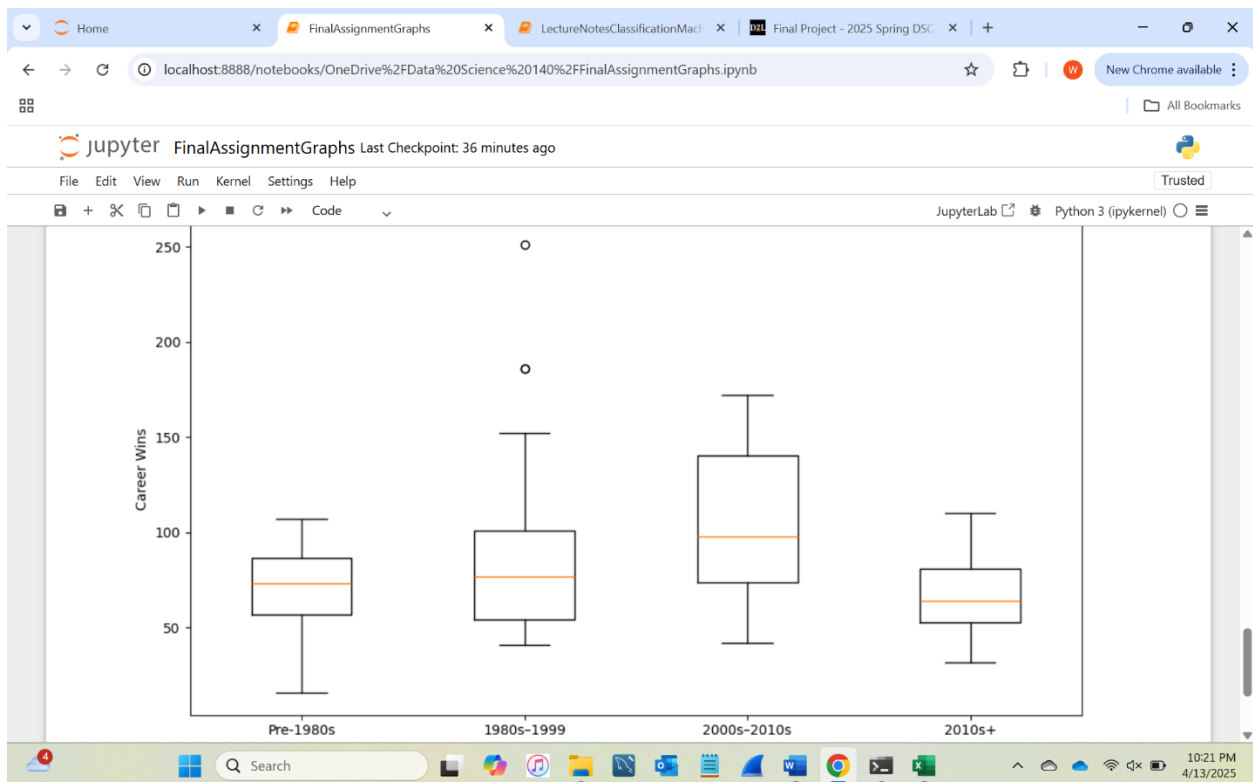
Figure 2: Interception/Touchdowns vs. Passer Rating



- Touchdowns and passer rating: $r \approx 0.79$
- Interceptions and passer rating: $r \approx -0.61$

This next graph shows the comparison of players, and their passer ratings compared to their interceptions and their touchdowns. This is because the two major stats that affect a player's passer rating are touchdowns (positive influence on passer rating) and interceptions (negative influence). What is seen clearly in both graphs is that there is no clear trend between the two stats and passer rating. You could say that the touchdowns graph has a slightly positive slope as touchdowns go up and that the interceptions have slightly negative slope but it's not too clear. This can be tossed up to the fact that there are many stats that affect someone's passer rating so once you get to the volume of statistics that these qb's have it can be very hard to decipher the true correlation.

Figure 3: Career Wins by Era



The final graph is a comparison of boxplots between the different eras of qb's and how many wins they have. It was decided to include the 2010's plus era but it will not be

discussed because they haven't had the time to get the volume of wins the other eras have. This is a very clear trend of qb's being more successful and more important to team success as time goes on that I expect to continue. So, the real question is why? Since before the 1980's people have wanted to watch qb's become more successful for entertainment reasons. They're the guys who get the ball every play, they make the big plays, they're the number one reason for team success, so the NFL has decided to help them out more and more. Rule changes benefitting the offense, style changes into the ideas of how offensive football should be played, and improvements to player development that specifically helps qb's more than every before is the main reason for this. Basically, qb is extremely important in football.

Discussion:

This project looked at the relationship between quarterback statistics and how successful their careers were, especially whether they made the Hall of Fame. The results showed that players with higher passing yards, more touchdowns, and more wins usually had better careers. There was a strong correlation between passing yards and wins, and between passing yards and touchdowns, which makes sense because good quarterbacks usually move the ball and help their teams win.

The KS test showed that passer rating alone didn't clearly separate Hall of Fame players from others. So, while a high passer rating is obviously good, it's not the only thing voters seem to care about. The Chi-Squared test showed that winning Super Bowls had a much stronger connection to getting into the Hall of Fame, meaning that championships matter a lot when it comes to a player's legacy.

The KNN machine learning model had about 83% accuracy, which is pretty good but not perfect. Some players were misclassified, suggesting that there are other things besides stats that matter, like leadership, team impact, or reputation. This shows that while stats tell a lot of the story, they don't tell everything.

Overall, the results mostly answered the original research questions. Performance stats and team success were important, but they weren't the whole story.

Conclusion:

In the end, the analysis showed that quarterbacks with more yards, touchdowns, wins, and especially Super Bowl wins were more likely to be seen as successful and to make the Hall of Fame. Stats like passer rating were helpful but not enough by themselves to explain everything. Using both stats and a machine learning model gave a better picture of what matters for a quarterback's career. Future studies could include other factors, like rushing yards, MVP awards, or playoff performances, to get an even deeper understanding of what leads to success.

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